

Customer Lifetime Value (CLTV) Prediction & Analysis Report

Machine Learning Hackathon Project

Project	CLTV Prediction Hackathon
Primary model	LightGBM Regressor
Target	Customer Lifetime Value (CLTV), modeled on a log scale
Evaluation metric	R ² score
Observed R ²	Approximately 0.13
Technology	Python, Pandas, NumPy, Matplotlib, scikit-learn, LightGBM

This report documents the CLTV modeling workflow, evaluation results, key observations, limitations, and recommended improvements.

1. Executive Summary

Customer Lifetime Value (CLTV) estimates the economic value a customer is expected to generate over a relationship with a business. The objective of this project was to build a regression model that predicts CLTV from customer-level features.

The workflow used Pandas and NumPy for data preparation, scikit-learn for model splitting and evaluation, Matplotlib for visualization, and LightGBM for regression. A logarithmic representation of CLTV was used to reduce the effect of extreme target values.

The current model achieved an R^2 score of approximately 0.13. This indicates that the model captures some predictive signal, but substantial variation in CLTV remains unexplained. The result should therefore be treated as a baseline rather than a production-ready model.

2. Problem Statement

Businesses need to identify customers who are likely to generate high long-term value. CLTV prediction can support customer segmentation, retention campaigns, marketing prioritization, and allocation of customer-acquisition or retention budgets.

Goal: Predict a numerical CLTV value from historical customer information and evaluate the predictive quality using a regression metric.

3. Project Structure

File / folder	Purpose
data/train.csv	Training data used for model development
data/test.csv	Test data used for prediction
notebooks/CLTV_Prediction.ipynb	Main analysis and modeling notebook
images/	Saved analysis and model visualizations
outputs/submission.csv	Prediction file prepared for submission
README.md	Project overview and documentation

4. Methodology

- **Data loading:** Training and test CSV files were loaded with Pandas.
- **Exploratory analysis:** The dataset was inspected for dimensions, feature types, target behavior, and useful relationships.
- **Target transformation:** CLTV was modeled on a logarithmic scale to reduce skew and the influence of extreme values.
- **Train/validation preparation:** scikit-learn's `train_test_split` utility was used for model development.
- **Model:** LightGBM Regressor was selected for nonlinear tabular regression.
- **Evaluation:** R^2 was used as the primary metric, supported by actual-vs-predicted visualization.

5. Model Evaluation

The model's predicted log(CLTV) values were compared with actual log(CLTV) values. The prediction-versus-actual plot showed predictions concentrated in a narrower band than the actual values. This behavior is consistent with the modest R^2 score.

Metric	Result	Interpretation
R^2	≈ 0.13	Some predictive signal is captured, but explanatory power is limited

Note: The R^2 value is approximately 0.13 based on the project result available during development. Replace it with the final notebook score if the finalized run produces a different value.

6. Key Observations

- The model learns a measurable relationship between the available features and CLTV.
- Predictions are concentrated around a narrower range than the actual target values, suggesting under-dispersion.
- The current feature set and/or model configuration does not explain enough of the variation in CLTV.
- The log target helps stabilize a skewed monetary target, but inverse transformation should be considered when interpreting predictions in the original CLTV scale.

7. Limitations

- The observed R^2 is relatively low.
- Feature quality and feature engineering may have a larger impact than simply increasing model complexity.
- A single train/validation split can be sensitive to the chosen split.
- The current result should not be interpreted as production-level performance or as a causal model.

8. Recommended Improvements

- **Feature engineering:** Add recency, frequency, monetary value, average order value, tenure, and customer interaction aggregates where available.
- **Outlier analysis:** Investigate extreme CLTV values and distinguish genuine high-value customers from data-quality issues.
- **Model comparison:** Compare LightGBM with Random Forest, XGBoost, CatBoost, and regularized linear models.
- **Hyperparameter tuning:** Tune learning rate, estimators, leaf/depth parameters, and sampling parameters using cross-validation.
- **Validation:** Use K-fold cross-validation or a time-aware split when temporal information is available.
- **Business evaluation:** Measure how well the model ranks the highest-value customers, not only its regression score.

9. Conclusion

This project demonstrates an end-to-end machine learning workflow for CLTV prediction, from data loading and preparation through modeling, evaluation, visualization, and Git/GitHub project organization. The LightGBM baseline shows that predictive signal exists, while the approximately 0.13 R^2 score highlights the need for stronger feature engineering, validation, and tuning before production use.

As a portfolio project, the work demonstrates practical experience with tabular data, regression modeling, visualization, model evaluation, reproducible project structure, and Git/GitHub workflow.